BSc (Hons) in Computer Systems Engineering

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Final Assignment

Design and Implementation of a 1-Digit Decimal Up Counter Using D Flip-Flops

IE2044 – System Modeling and Prototyping

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1.Introduction

This project focuses on designing and implementing a 1-digit decimal (0–9) up counter using D flip-flop ICs (74HC74) in a ripple (asynchronous) counter configuration. The counter automatically increments its value from 0 to 9 and then resets back to 0, displaying the count on a seven-segment display through a BCD to 7-segment decoder. Additional control features such as Pause/Resume and Reset are implemented using push buttons to manually control the counting process. The complete circuit is designed, simulated, and verified in Proteus**,** then converted into a PCB layout using EasyEDA. Finally, a simple enclosure is designed to safely house the finished PCB and components, ensuring a functional and professional final product.

2.Design Methodology

The design process of the 1 digit decimal up counter was divided into several key stages to ensure systematic development, correct functionality, and reliable hardware implementation.

2.1 Circuit Design

- The circuit was designed using D flip-flop ICs (74HC74) configured in a ripple (asynchronous) counter arrangement. The counter counts binary values from 0000 (0) to 1001 (9). A pulse generator provides the clock signal to drive the counter. The NAND gate (74HC00) detects the binary count ‘1010’ (decimal 10) and generates a reset pulse to clear the counter back to zero, achieving a mod-10 counter**.** The four output bits (Q0–Q3) are connected to a BCD-to-7-segment decoder (74LS48)**,** which drives a common-anode 7-segment display (FJS161A) to show decimal digits 0–9. Two push buttons are included one for asynchronous reset and another for pause/resume control of the counter.

2.2 Functional Blocks

- Clock Input: Provides a continuous pulse to drive the counter. It can be generated using a timer circuit or an external clock signal.
- Counter Section: Two 74HC74 dual D flip-flop ICs are cascaded to form a 4-bit asynchronous counter, with each flip-flop toggling on the falling edge of the previous stage.
- Reset Logic: A NAND gate (74HC00) detects the count 1010 (decimal 10) and outputs a low signal to reset all flip-flops, ensuring the counter operates within the 0–9 range.
- Display Driver: The 74LS48 decoder converts the binary count into signals suitable for driving the 7-segment display, enabling clear visualization of each digit.
- Push Buttons:
    - Reset Button: Clears the count to 0 asynchronously.
    - Pause/Resume Button: Temporarily disables or resumes the clock signal, allowing manual control over counting.

2.3 Operating Principle

- The first D flip-flop toggles its output with every clock pulse, while each subsequent flip-flop is triggered by the output of the previous stage, creating a ripple counting sequence. The binary count progresses from 0000 (0) to 1001 (9). When the binary value 1010 (10) occurs, the NAND gate output becomes low, resetting all flip-flops to 0000, thereby restarting the counting cycle. The binary outputs (Q0–Q3) are fed to the 74LS48 decoder, which drives the 7-segment display to show the corresponding decimal digit. The pause/resume control allows the user to freeze or continue the counting process without affecting the stored count.

2.4 PCB Design

- After the schematic was verified through simulation, the design was transferred to EasyEDA for PCB layout creation. Proper component placement, trace routing, and ground plane design were ensured to minimize noise and interference. Signal labels such as CLK, RESET, VCC, and GND were added for clarity. Once finalized, the PCB design was checked using the built-in Design Rule Check (DRC) tool and prepared for fabrication.

3. Component Selection

Component	Part Number / Code	Quantity
D-Type Flip-Flop IC	74HC74	2
NAND Gate IC	74HC00	1
AND Gate IC	74HC08	1
BCD to 7 Segment Decoder	74LS47	1
7-Segment Display (Common-Anode)	FJS161A	1
Resistors (330Ω, 1kΩ, 10kΩ, 47kΩ)	Various	8+
Capacitors (10µF, 22nF)	Various	2
Pushbuttons	Generic	2
Header Pins	Generic	3

Each component in the circuit was carefully selected to ensure correct functionality, compatibility, and reliable performance:

- All components were chosen based on their availability and compatibility with TTL logic levels.
- ICs from the 74-series family ensure reliable and fast switching performance.
- Resistors and capacitors are used for pull-up, timing, and debounce functions.
- The 7-segment display and BCD decoder simplify visual output representation.
- Pushbuttons are used for manual clock and reset inputs.
- Header pins allow easy connections for testing and interfacing in Proteus.

4. Results & Testing

- The circuit was first tested and verified using Proteus simulation, and its actual physical layout was implemented in EasyEDA.

4.1 Proteus Simulation

- The complete circuit was constructed and simulated in Proteus to confirm the counting logic, reset functionality, and display output.

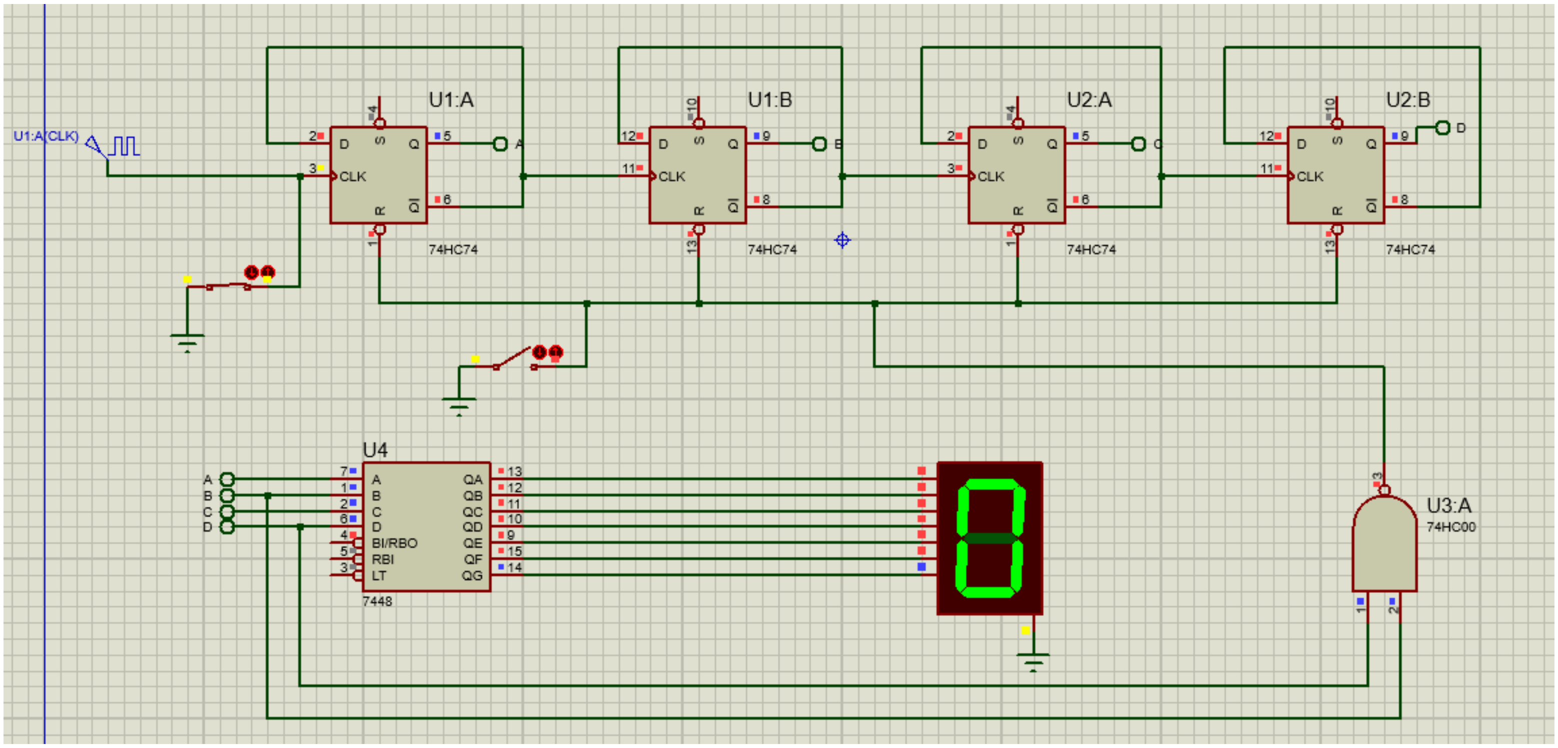

Figure 1 Complete Circuit Schematic in Proteus

Simulation Results:

- The simulation confirmed that the counter correctly cycles from 0 through 9 before resetting to 0. The pause and reset buttons were also tested and found to be fully functional.

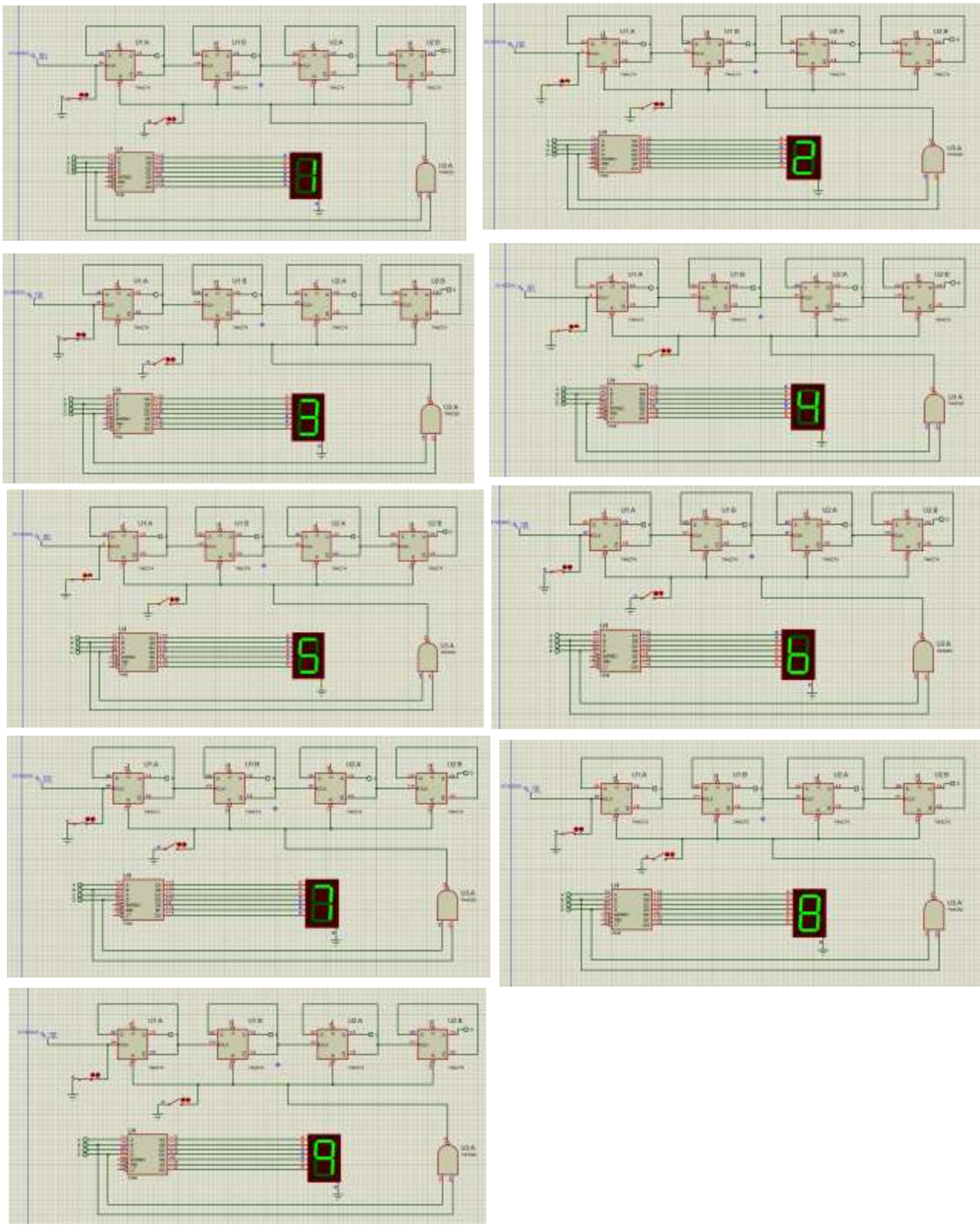

Figure 2 Simulation showing the counter displaying various digits during operation.

4.2 EasyEDA PCB Design

After verification, the schematic was transferred to EasyEDA to create the final PCB layout.

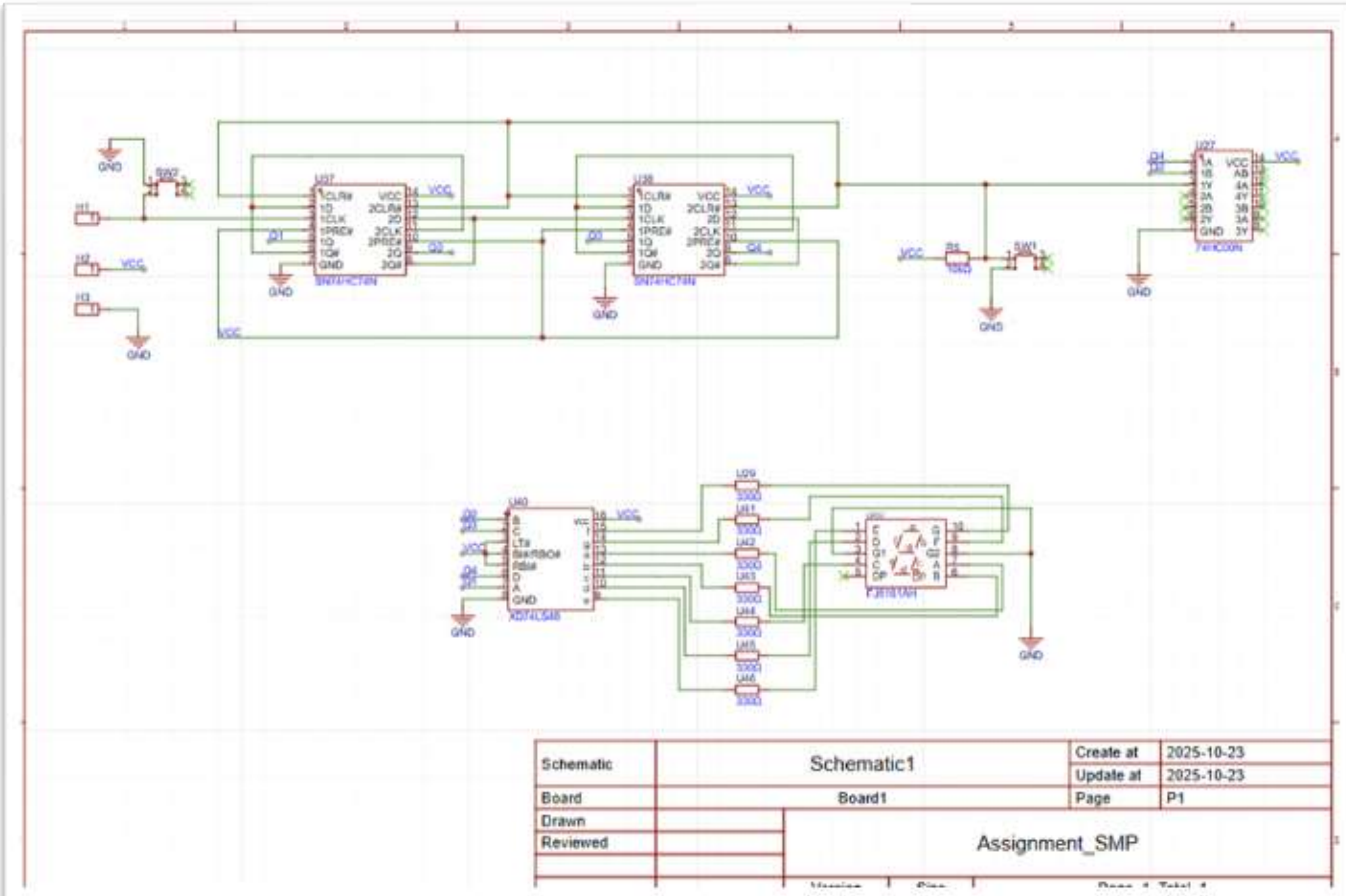

Figure 3 Final Circuit Schematic in EasyEDA

- The D flip-flops (74HC74) form the core of the ripple counter.
- A BCD to 7-segment decoder (74LS47) is used to drive the display.
- Reset and Pause/Resume buttons are included for user control.
- All components were labeled and verified before generating the PCB layout.
- Proper VCC and GND connections were ensured to maintain signal stability.

4.3 PCB Layout

- Component placement was optimized to group related functional blocks and minimize trace lengths. Power and ground traces were designed with increased width to reduce voltage drops. The top layer (red) primarily carries signal traces, while the bottom layer (blue) serves as a ground plane to enhance overall circuit stability and noise performance.

PCB Layout – Top layer

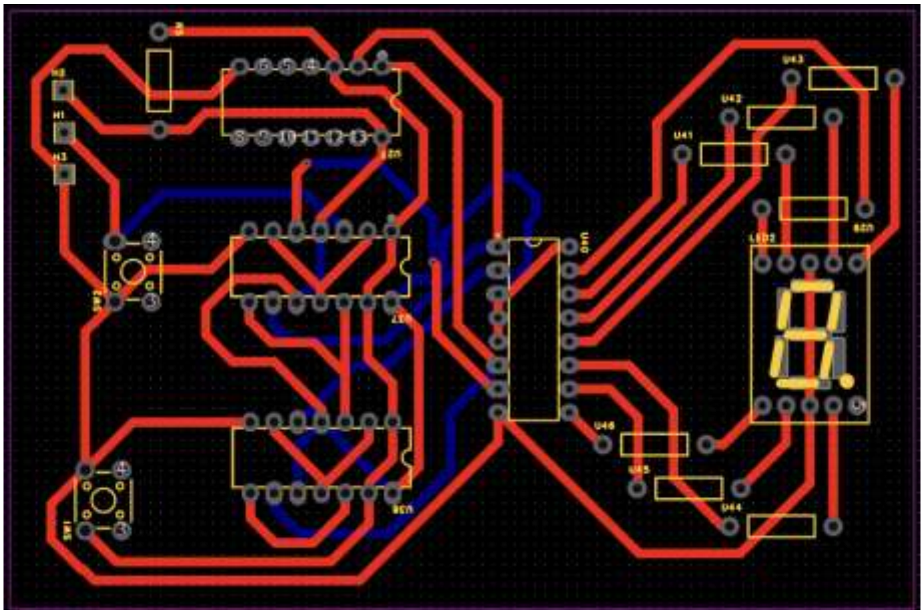

Figure 4 PCB Layout - Top Layer

PCB Layout – Bottom layer

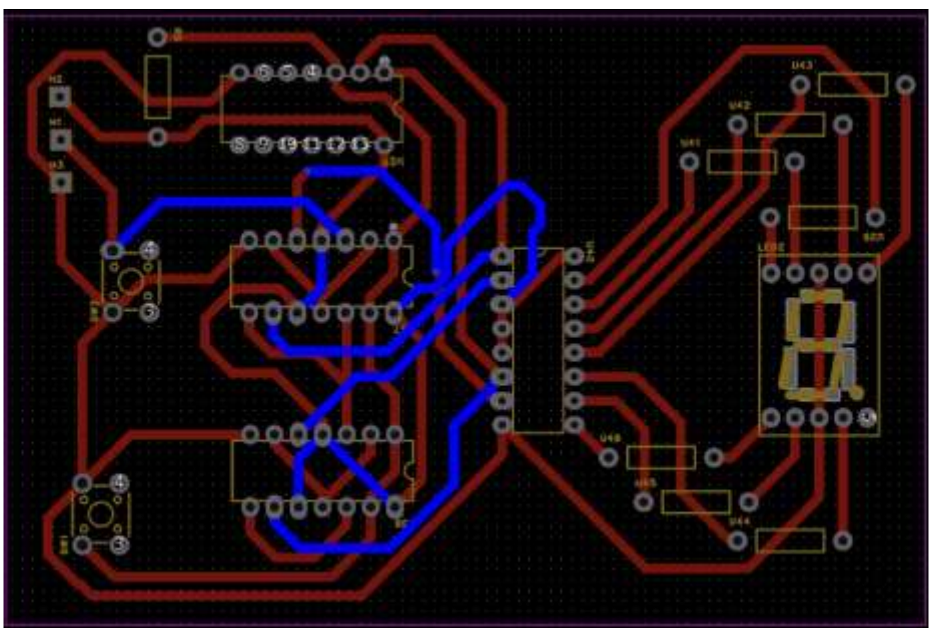

Figure 5 PCB Layout - Bottom Layer

4.4 2D PCB Design

- EasyEDA's 2D viewer was utilized to inspect and visualize the final assembled PCB layout.

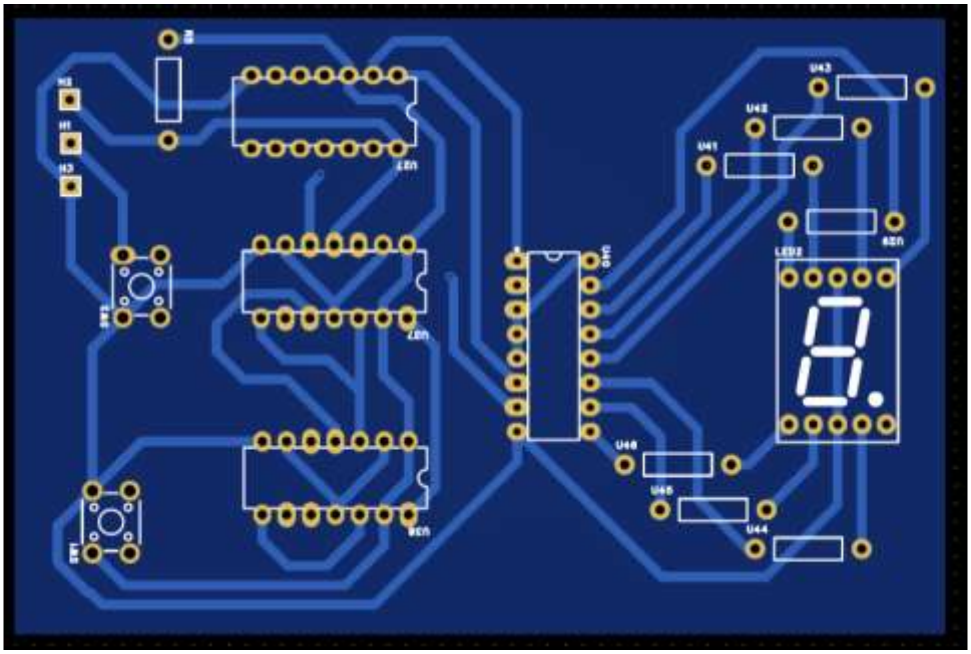

Figure 6 2D Model of the Final PCB Design

4.5 3D PCB Design

- Functional blocks were grouped for clarity and minimal trace length.
- Power and ground traces were widened for stability.
- Top layer handles signal routing: bottom layer serves as ground plane.

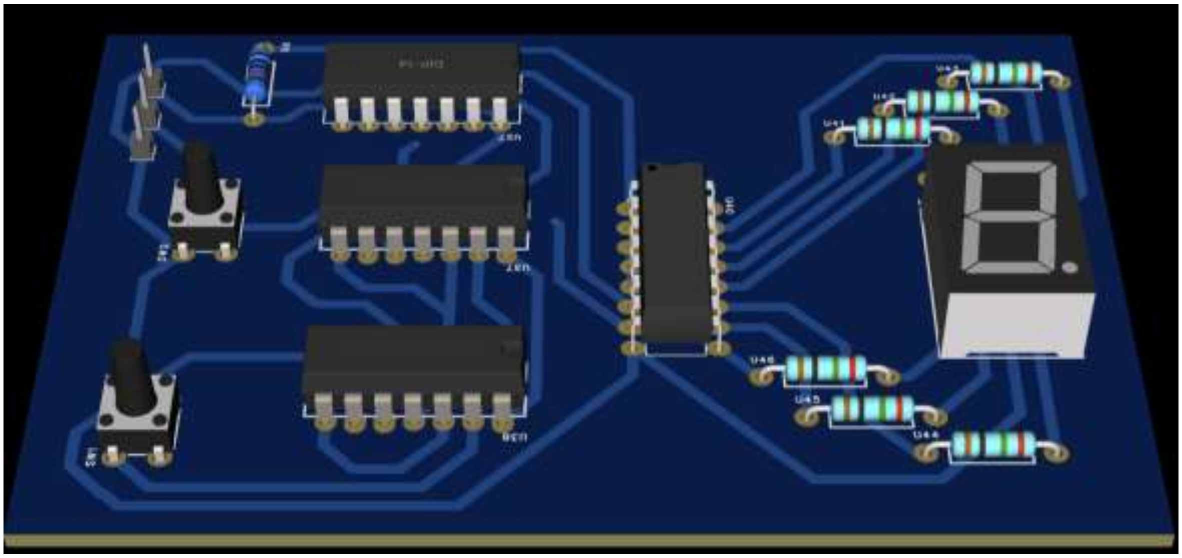

Figure 7 3D Model of the Final PCB Design

4.5 Enclosure Design

- Autodesk Fusion was used to model a two-part enclosure.
- Lid includes cutouts for display and buttons.
- Base includes mounting posts for PCB stability.

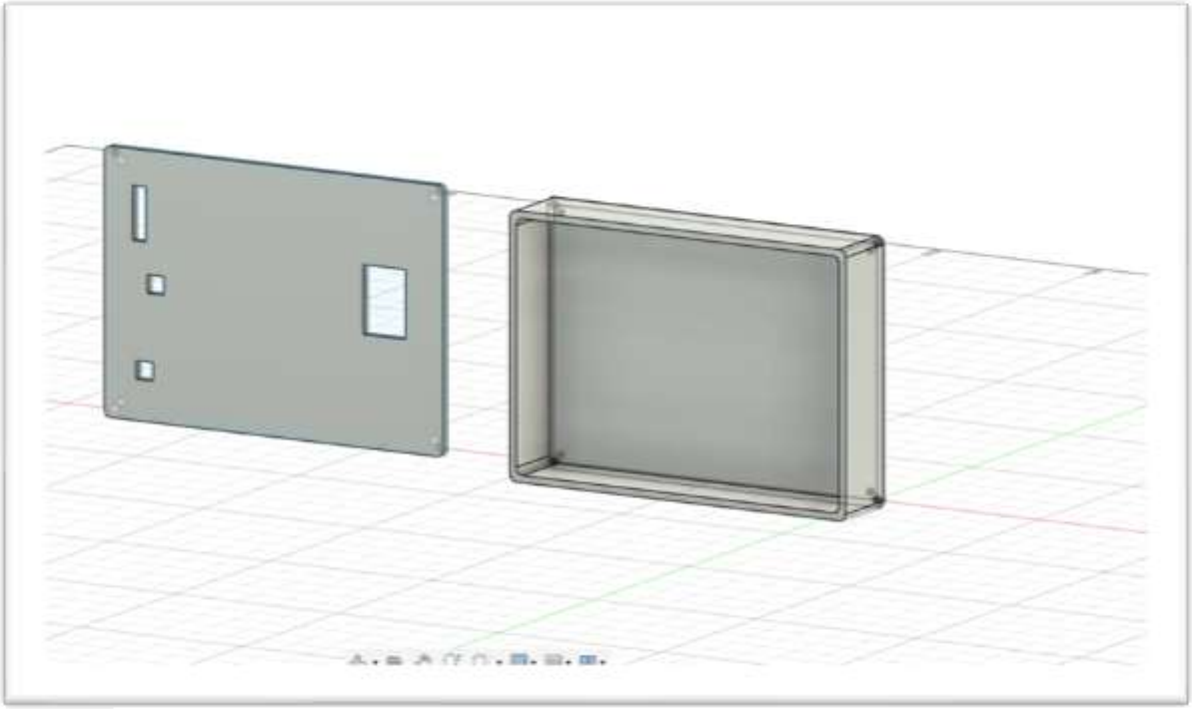

Figure 8 3D Model of the Enclosure Base and 3D Model of the Enclosure Lid

5. Discussion

- The project provided valuable hands-on experience in digital circuit design and practical PCB implementation.

- A key challenge was adapting the reset logic from active-high to active-low due to changes in flip-flop ICs between simulation and final hardware.

- This emphasized the importance of thoroughly reading datasheets and understanding pin configurations before circuit assembly.

- The need to terminate all unused CMOS inputs was reinforced to avoid instability caused by floating pins.

- Component placement and routing were carefully optimized to reduce interference and signal delay.

- The design demonstrated how propagation delays can affect asynchronous counters and influence output accuracy.

- Testing and troubleshooting improved skills in logic analysis, circuit debugging, and waveform interpretation.

- Simulation results closely matched the practical outcomes, confirming correct logic design and hardware implementation.

- The experience highlighted the importance of grounding, decoupling capacitors, and proper power distribution for stable circuit performance.

- Although the asynchronous design met the requirements, a synchronous counter is recommended for future improvements to ensure better timing and reliability.

6. Conclusion

The project successfully met all objectives. A fully functional 1-digit decimal counter was designed, simulated, and prepared for manufacturing. The integration of digital logic, PCB design, and enclosure modeling demonstrated a strong understanding of system design and prototyping. The process improved skills in circuit design, simulation, and troubleshooting, while emphasizing the importance of proper component selection and datasheet analysis. Future improvements could include developing a multi-digit or synchronous version to enhance speed and reliability.

7.Reference

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